

Aaron Rapp, Ph.D.

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Employment History

- 2025 – Present ■ **Assistant Professor of Mathematics**, Winston Salem State University.
- 2020 – 2025 ■ **Assistant Professor of Mathematics**, University of the Virgin Islands.
- 2022 – 2025 ■ **Data Science Program Coordinator**, University of the Virgin Islands.
- 2017 – 2019 ■ **Lecturer**, Residential Colleges, The University of North Carolina at Greensboro.
- 2016 – 2020 ■ **Graduate Teaching Assistant/Associate**, Department of Mathematics and Statistics, The University of North Carolina at Greensboro.
- 2019 ■ **SOAR Advisor**, New Student Transitions & First Year Experience, The University of North Carolina at Greensboro.
- 2016 ■ **Underwriting in Advance Analytics Intern**, AIG-United Guarantee Corporation.
- 2013 – 2015 ■ **Graduate Teaching Assistant**, College of Arts and Sciences, Western Carolina University.
- 2014 – 2015 ■ **Summer Lecturer**, First Year Experience Program, Western Carolina University.

Education

- 2015 – 2020 ■ **Ph.D. in Computational Mathematics**, The University of North Carolina at Greensboro.
Dissertation title: *Symmetric dual-wind discontinuous Galerkin methods for elliptic variational inequalities.*
- 2013 – 2015 ■ **M.S. in Applied Mathematics** Western Carolina University.
Thesis title: *A Lifting of Graphs to 3-Uniform Hypergraphs, Its Generalization, and Further Investigation of Hypergraph Ramsey Numbers.*
- 2009 – 2013 ■ **B.S. in Mathematics** Mars Hill University.

Research Interests

- **Primary Interests** Biomedical and Public Health Data Science, Numerical Solutions to PDEs, Finite Element Methods, Discontinuous Galerkin Methods.
- **Application Areas:** Electronic Health Record Analysis, Machine Learning Analysis, Deterministic and Stochastic Optimal Control, Optimal Mass Transport, Materials Science, Complex Fluid, Mathematical Biology, Machine Learning integrated with PDE approximation methods.
- **Education:** Impact of Open Education Resources, Social Justice in STEM

Academic Works

Journal Articles

- 1 S. B. Boyana, T. Lewis, A. Rapp, and Y. Zhang, "Convergence analysis of a symmetric dual-wind discontinuous Galerkin method for a parabolic variational inequality," *Journal of Computational and Applied Mathematics*, vol. 422, p. 114 922, 2023.
- 2 T. Lewis, A. Rapp, and Y. Zhang, "Consistency results for the dual-wind discontinuous Galerkin method," *Journal of Computational and Applied Mathematics*, vol. 431, p. 115 257, 2023.
- 3 X. Feng, T. Lewis, and A. Rapp, "Dual-wind discontinuous Galerkin methods for stationary Hamilton-Jacobi equations and regularized Hamilton-Jacobi equations," *Communications on Applied Mathematics and Computation*, vol. 4, no. 2, pp. 563–596, 2022.
- 4 T. Lewis, A. Rapp, and Y. Zhang, "Penalty parameter and dual-wind discontinuous Galerkin approximation methods for elliptic second orders PDEs," *Electronic Journal of Differential Equations*, vol. Conference 26, pp. 123–138, 2022.
- 5 T. Lewis, A. Rapp, and Y. Zhang, "Convergence analysis of symmetric dual-wind discontinuous Galerkin approximation methods for the obstacle problem," *J Math Anal Appl*, vol. 485, no. 2, p. 123 840, 2020.
- 6 A. Rapp, "Symmetric dual-wind discontinuous Galerkin methods for elliptic variational inequalities," Ph.D. dissertation, 2020.
- 7 M. Budden, J. Hiller, and A. Rapp, "Hypergraph Ramsey numbers involving paths," *Acta Univ. Apulensis*, vol. 48, pp. 75–87, 2016.
- 8 M. Budden, J. Hiller, and A. Rapp, "Generalized Ramsey theorems for r-uniform hypergraphs.," *Australas. J Comb.*, vol. 63, pp. 142–152, 2015.
- 9 M. Budden and A. Rapp, "Constructing r-uniform hypergraphs with restricted clique numbers," *The North Carolina Journal of Mathematics and Statistics*, vol. 1, pp. 30–34, 2015.
- 10 A. Rapp, "A lifting of graphs to 3-uniform hypergraphs, its generalization, and further investigation of hypergraph Ramsey numbers," M.S. thesis, Western Carolina University, 2015.

Invited Presentations

- 1 A. Rapp, J. Monestime, and P. Alexandre, "When policy isn't enough: Explaining state variation in medicaid providers' transition to interoperability," *AcademyHealth Annual Reserach Meeting - State Health Policy and Research Interest Group Meeting*, Jun. 2026, Seattle Convention Center.
- 2 A. Rapp, L. Zhang, S. Nduaguba, and S. Hasan, "Impact of endocrine system failures in long Covid diagnoses: Analyzing the effects of diabetes, hypo or hyperthyroidism, and adrenal insufficiency," *Introduction to Translational Team Project Course*, Sep. 2024, NCATS and AIM-AHEAD Advanced Data Analysis Program.
- 3 A. Rapp, "Approximating the solutions to Hamilton-Jacobi equations with dual-wind discontinuous Galerkin methods," *The 13th AIMS Confernece on Dynamical Systems, Differential Equations and Applications, Mini-Symposium on Recent Advancements in the Numerical Analysis of Nonlinear Partial Differential Equations*, Jun. 2023, University of North Carolina at Wilmington.
- 4 A. Rapp, "The dual-wind discontinuous Galerkin method for Hamilton-Jacobi equations," *10th International Congress on Industrial and Applied Mathematics, Mini-symposium on Innovative Numerical Methods for Complex PDEs*, Aug. 2023, Waseda University.

- 5 A. Rapp, "Consistency and penalty parameter results for the dual-wind discontinuous Galerkin methods," *2nd Annual International Conference on Computational Methods and Applications, Mini-Symposium on Recent Developments in Finite Difference and Finite Element Methods*, May 2022, Mississippi State University.
- 6 A. Rapp, "Dual-wind discontinuous Galerkin methods for Hamilton-Jacobi equations," *AMS Fall Central Section Meeting, Special Session on Recent Advances in Scientific Computing and Application*, Sep. 2022, University of Texas at El Paso.
- 7 A. Rapp, "Dual-wind discontinuous Galerkin methods for time-dependent Hamilton-Jacobi equations," *The 7th Annual Meeting of SIAM Central States Section, Mini-Symposium on New Trends on Discontinuous Galerkin Methods for Partial Differential Equations*, Oct. 2022, Oklahoma State University.
- 8 A. Rapp, "The consistency of the dual-wind discontinuous Galerkin methods," *UNC Greensboro Partial Differential Equation Conference*, Jul. 2021, Virtual Conference.
- 9 A. Rapp, "(cancelled due to covid-19) the dual-wind discontinuous Galerkin method for the obstacle problem and its optimal control variation," *Numerical Methods for Optimization Problems with PDE Constraints Mini-Symposium at the International Conference on Computational Methods and Applications in Engineering*, May 2020, Mississippi State University.
- 10 A. Rapp, "Convergence analysis of symmetric dual-wind discontinuous Galerkin methods for the obstacle problem," *Joint Mathematics Meetings AMS Special Session on Recent Developments in Numerical Methods for PDEs*, Jan. 2020, Denver, CO.
- 11 A. Rapp, "Dual-wind discontinuous Galerkin method with applications to obstacle problems," *Finite Element Circuits*, Nov. 2019, Virginia Tech.
- 12 A. Rapp, "Dual-wind discontinuous Galerkin method with applications to obstacle problems," *The American Women in Mathematics Piedmont-Triad Conference*, Mar. 2019, Wake Forest University.
- 13 A. Rapp, "Optimization of the penalty parameter for the dual-wind discontinuous Galerkin methods on a prototypical second order pde," *39th Southeastern-Atlantic Regional Conference on Differential Equations (SEARCDE)*, Oct. 2019, Embry-Riddle Aeronautical University.
- 14 A. Rapp, "Dual-wind discontinuous Galerkin method with applications to obstacle problems," *The 14th Annual UNCG Regional Mathematics and Statistics Conference*, Nov. 2018, The University of North Carolina at Greensboro.
- 15 A. Rapp, "Dual-wind discontinuous Galerkin methods for elliptic obstacle problem," *38th Southeastern-Atlantic Regional Conference on Differential Equations (SEARCDE)*, Oct. 2018, University of North Georgia.
- 16 A. Rapp, "Constructing r -uniform hypergraphs with restricted clique sizes," *The 10th Annual UNCG Regional Mathematics and Statistics Conference*, Nov. 2014, The University of North Carolina at Greensboro.

Poster Presentations

- 1 A. Rapp and J. Monestime, "State variation in provider progression to modified stage 2 meaningful use: A comparative study of north carolina and florida medicaid interoperability programs," *Conference on Health IT and Analytics*, May 2026, John Hopkins University.
- 2 A. Rapp, J. Monestime, and P. Alexandre, "Markers of success in modified stage 2 meaningful use: Implications for medicaid policy," *AcademyHealth Annual Research Meeting*, Jun. 2026, Seattle Convention Center.
- 3 A. Rapp, A. Bellew, J. Orr, T. Petika, and C. Harrison, "Advancing health in the united states virgin islands through ai/ml models by utilizing electronic health records in the aim-ahead federated network," *AIM-AHEAD 2025 Annual Meeting*, Jul. 2025, Addison, TX.

- 4 A. Rapp, L. Zhang, and S. Nduaguba, "Impact of endocrine system failures in long Covid diagnoses: Analyzing the effects of diabetes, hypo or hyperthyroidism, and adrenal insufficiency," *AIM-AHEAD AI for Health Equity Symposium*, Aug. 2024, Atlanta, GA.

Undergraduate Mentee Presentations

- 1 A. Rapp, J. Monestime, and A. Edgerton, "The 2011-2021 north carolina medicaid interoperability program: Understanding the characteristics of eligible providers who achieved modified stage 2 meaningful use," *WSSU Scholarship Day*, Apr. 2026, Winston-Salem State University.
- 2 L. Alexander and M. Roberts, "Understanding the readmittance rates of diabetic patients through random forest models," *25th Annual University of the Virgin Islands Undergraduate Reserach Symposium*, 2024, St. Thomas, USVI.
- 3 J. Buchanan, Z. Byron, and M. Todman, "Determining important factors of readmittance rates of diabetic patients using xgboost and support vector machine models," *25th Annual University of the Virgin Islands Undergraduate Reserach Symposium*, 2024, St. Thomas, USVI.
- 4 A. Mora, "Ranger bot research," *24th Annual University of the Virgin Islands Undergraduate Reserach Symposium*, 2023, St. Thomas, USVI.
- 5 K. Rodgers and J. Wyllis, "Modeling the covid-19 epidemic in the USVI," *23rd Annual University of the Virgin Islands Undergraduate Reserach Symposium*, 2022, St. Thomas, USVI.
- 6 K. Vanterpool, "Approximating the heat equation using coding," *22nd Annual University of the Virgin Islands Undergraduate Reserach Symposium*, 2021, St. Thomas, USVI.

Professional Development Presentations

- 1 A. Rapp, "A crash course in data visualization," *The Undergraduate Research Training Initiative for Student Enhancement (U-RISE) Program*, Jan. 2024, The University of the Virgin Islands.
- 2 A. Rapp, "A crash course in data visualization," *Emerging Caribbean Scientists Program*, May 2024, The University of the Virgin Islands.
- 3 A. Rapp, M. Boumedine, and M.-J. Deans, "Embracing AI at UVI," *Faculty Professional Development Day*, May 2024, The University of the Virgin Islands.
- 4 A. Rapp, "A crash course in data visualization," *The Undergraduate Research Training Initiative for Student Enhancement (U-RISE) Program*, Feb. 2023, The University of the Virgin Islands.
- 5 A. Rapp, "Approximating solutions of partial differential equations with a discontinuous method.," *UVI Mathematics Junior and Senior Seminar Course*, Feb. 2023, The University of the Virgin Islands.
- 6 A. Rapp, "Technology talks - AI and ML opportunities," *AIM-AHEAD Southeast Region Hub*, Jun. 2023, The University of the Virgin Islands.
- 7 A. Rapp, "What does AI mean for UVI?" *Faculty Roundtable Discussion*, Mar. 2023, The University of the Virgin Islands.
- 8 A. Rapp, "An introduction to numerical analysis research," *UVI Mathematics Junior and Senior Seminar Course*, Feb. 2022, The University of the Virgin Islands.
- 9 A. Rapp, "Learning to use LaTeX," *UVI Mathematics Junior and Senior Seminar Course*, Aug. 2022, The University of the Virgin Islands.
- 10 A. Rapp, "What is statistics?" *Emerging Caribbean Scientists Program*, Jun. 2022, The University of the Virgin Islands.

Community Presentations

- 1 A. Rapp, M. Boumedine, M. Roberts, and L. Alexander, "Demonstrating the biomedical data science research conducted by UVI undergraduate students and faculty," *Biomedical Building Community Soft Opening*, Sep. 2024, The University of the Virgin Islands.
- 2 A. Rapp, M. Boumedine, M. Roberts, and L. Alexander, "Demonstrating the biomedical data science research conducted by UVI undergraduate students and faculty," *Biomedical Building RTPark Donor Event*, Oct. 2024, The University of the Virgin Islands.
- 3 A. Rapp, M. Boumedine, M. Roberts, and L. Alexander, "Demonstrating the biomedical research conducted by UVI undergraduate students," *Biomedical Building Soft Opening*, Aug. 2024, The University of the Virgin Islands.

Grants (PI or Co-PI)

2026	■	<i>Southeastern States Medicaid Interoperability: Understanding the Characteristics of Eligible Professionals Who Achieved Meaningful Use to Inform Future Medicaid Policy</i> , WSSU Startup Funding Program, Jan 2026 - June 2026, (\$6,000).
2024 – Present	■	<i>Advancing Health Equity and Inclusion in the United States Virgin Islands: The Utilization of Electronic Health Records in the AIM-AHEAD Federated Network to Support AI/ML Models for Underrepresented Populations</i> , Artificial Intelligence/Machine Learning Consortium to Advance Health Equity and Researcher Diversity (AIM-AHEAD) Program, Oct 2024- Sept 2026, (\$387,388).
2025 – Present	■	<i>Establishment of the U.S. Virgin Islands Center for Autonomous Research for a new Era of Ecological Marine Science and Workforce Development</i> , OIA - EPSCoR RISE RII, July 2025 - June 2029, (\$7,000,000).
2022 – 2025	■	<i>Increasing the Reach and Effectiveness of the UVI Data Science Program</i> . U.S. Department of Education Title III Part B, Oct 2022- Sept 2027 (\$513,562).
2022	■	<i>A New Approximation Method for Hamilton-Jacobi Equations</i> , VI NASA EPSCoR Research Initiation Award, May - Aug 2022 (\$20,000).

Reviewer

2025	■	HDSTP Health Data Analysis - Alumni Mentor
	■	NIH AIM-AHEAD Year 4 (2025-2026) Research Fellowship Program
2024	■	NIH AIM-AHEAD Year 3 (2024-2025) Research Fellowship Program
	■	25th Annual University of the Virgin Islands Undergraduate Research Symposium
2023	■	Journal of Mathematics
	■	The North Carolina Journal of Mathematics and Statistics
	■	24th Annual University of the Virgin Islands Undergraduate Research Symposium
2022	■	23rd Annual University of the Virgin Islands Undergraduate Research Symposium
2021	■	22nd Annual University of the Virgin Islands Undergraduate Research Symposium

Miscellaneous Experience

Awards and Achievements

- 2025 ■ **Best Poster Presentation Award**, AIM-AHEAD Annual Meeting 2026.
- 2024 ■ **Marshall**, University of the Virgin Islands 2024 Spring Commencement Ceremony.
- **Trainee**, NIH AIM-AHEAD and NCATs Traineeship in Advanced Data Analysis.

Miscellaneous Experience (continued)

- 2023  **Participant**, NIH Engagement and Access for Research-Active Institutions (EARA) Pilot.
- 2023  **Co-Organizer**, Innovative Numerical Methods for Complex PDEs Mini-Symposium at the 10th International Congress on Industrial and Applied Mathematics.
- 2023  **SIAM Early Career Travel Award**, The Society for Industrial and Applied Mathematics.
- 2020  **Project NExT Fellow**, Mathematical Association of America.
- 2020  **Graduate Student Travel Grant**, American Mathematical Society.
- 2018-2020  **Lula Martin McIver Fellowship (Formerly the Inclusiveness Award)**, The University of North Carolina at Greensboro Graduate School.
- 2017-2020  **Helen Barton Scholarship**, The University of North Carolina at Greensboro Mathematics Department.
- 2018  **Outstanding Graduate Student Presentation**, The 14th Annual UNCG Regional Mathematics and Statistics Conference.
- 2018  **Dr. Theresa Phillips Vaughn Math Scholarship**, The University of North Carolina at Greensboro Mathematics Department.
- 2016  **Fellow**, Opportunity Greensboro: A Business & Collegiate Partnership.

Certifications

- 2024  **CF-L2**. Cross-Fit, LLC.
- 2022  **Online Teaching - Blackboard (16 CEU)**. Center for Excellence in Teaching and Learning at the University of the Virgin Islands.
- 2021  **Course Development - Blackboard (20 CEU)**. Center for Excellence in Teaching and Learning at the University of the Virgin Islands.

Secondary Employment

- 2022-Present  **Cross-Fit Instructor**. Rock Life Cross-Fit.
- 2023-2024  **Statistical Analyst** Augmented Reality Personalized Motion Tracking Assessment and Improvement Technology for Fall Prevention for the Spanish Speaking Hispanic/Latino Population in a Community Setting Using Mobile Phone (NIH Award Number 1R43AG080853-01).

Committee Membership

- 2025-2026  **Chair of the Introduction to Technical Computing Course Development Committee**. Winston-Salem State University.
- 2025-Present  **Academic Standards and Curriculum Committee**. Winston-Salem State University.
- 2024-2025  **Universal Data Warehouse Leadership Committee**. The U.S. Virgin Islands Department of Health.
- 2023-2025  **American Association of University Professors Executive Committee**. The University of the Virgin Islands.
- 2022-2025  **Emerging Caribbean Scientists Steering Committee**. University of the Virgin Islands.
- 2024  **Title III Program Assistant Search Committee** University of the Virgin Islands.
- 2023  **Biology Department Search Committee** University of the Virgin Islands.

Miscellaneous Experience (continued)

	📖	Applied Mathematics Degree Committee University of the Virgin Islands.
2020-2023	📖	English Proficiency Exam Committee University of the Virgin Islands.
2021, 2022, 2023	📖	Math Department Faculty Search Committee(s) University of the Virgin Islands.

Organizational Membership

2026-Present	📖	Academy Health
2023-Present	📖	The Society for Industrial and Applied Mathematics.
2020-2022	📖	Mathematical Association of America.
2015-2020	📖	American Mathematical Society.
2016-2020	📖	UNCG American Mathematical Society Graduate Student Chapter (Founder).
2018-2020	📖	Promoting Inclusion in Science, Technology, Engineering and Mathematics.
2019-2020	📖	Association for Women in Mathematics.

Courses Taught

Winston-Salem State Univeristy

Spring 2026	📖	MAT 3419-01 - Calculus III
	📖	MAT 4311-01 - Real Variables
Fall 2025	📖	MAT 2714-01 - Calculus 1
	📖	MAT 1311-03 - College Algebra
	📖	MAT 1311-04 - College Algebra

University of the Virgin Islands

Spring 2025	📖	MAT 342 - Intermediate Calculus II
	📖	CSC/IST 435 - Data Science II
	📖	MAT 245 - Statistics for the Life Sciences
Fall 2024	📖	MAT 341 - Intermediate Calculus I
	📖	MAT 325 - Numerical Analysis
	📖	MAT 233 - Discrete Mathematics
Summer 2024	📖	MAT 153 - College Trigonometry
Spring 2024	📖	CSC/IST 435 - Data Science II
	📖	MAT 346 - Differential Equations
	📖	MAT 245 - Statistics for the Life Sciences
Fall 2023	📖	MAT 362 - Abstract Algebra
	📖	MAT 261 - Linear Algebra
	📖	MAT 233 - Discrete Mathematics
Summer 2023	📖	MAT 153 - College Trigonometry
Spring 2023	📖	CSC/IST 435 - Data Science II
	📖	MAT 245 - Statistics for the Life Sciences
Fall 2022	📖	CSC/IST 230 - Data Science I
	📖	MAT 261 - Linear Algebra
	📖	MAT 233 - Discrete Mathematics

Courses Taught (continued)

Summer 2022	📖	MAT 153 - College Trigonometry
Spring 2022	📖	MAT 346 - Differential Equations
	📖	MAT 342 - Intermediate Calculus II
	📖	MAT 245 - Statistics for the Life Sciences
Fall 2021	📖	MAT 362 - Abstract Algebra
	📖	MAT 233 - Discrete Mathematics
	📖	MAT 153 - College Trigonometry
Summer 2021	📖	MAT 143 - Pre-Calculus Algebra
Spring 2021	📖	MAT 346 - Differential Equations
	📖	MAT 241 - Calculus II
Fall 2020	📖	MAT 261 - Linear Algebra Algebra
	📖	MAT 241 - Calculus I
	📖	MAT 140 - College Algebra

The University of the North Carolina at Greensboro

Spring 2020	📖	MAT 190 - Pre-Calculus Algebra
Fall 2019	📖	MAT 191 - Calculus I (2 Sections)
Spring 2019	📖	MAT 112 - Contemporary Topics in Mathematics (2 Sections)
	📖	RCO 115 - College Algebra
Fall 2018	📖	MAT 191 - Calculus I
	📖	MAT 120 - Calculus with Business Applications
	📖	RCO 115 - College Algebra
Spring 2018	📖	MAT 115 - College Algebra
	📖	RCO 115 - College Algebra
	📖	MAT 112 - Contemporary Topics in Mathematics
Fall 2017	📖	RCO 115 - College Algebra
	📖	MAT 112 - Contemporary Topics in Mathematics (2 Sections)
Spring 2017	📖	MAT 112 - Contemporary Topics in Mathematics (1 In-Person and 2 Online Sections)
Fall 2016	📖	MAT 112 - Contemporary Topics in Mathematics (2 Online Sections)
	📖	MAT 151 - Pre-Calculus II

Western Carolina University

Summer 2015	📖	USI 130 - The University Experience
Spring 2015	📖	MATH 170 - Applied Statistics
Fall 2014	📖	MATH 170 - Applied Statistics
Summer 2014	📖	USI 130 - The University Experience

Skills

Programming Languages	📖	Python, C++, Java, Android
Mathematical Software	📖	R, MATLAB, $\text{\LaTeX}$, Mathematica, MAGMA

Relevant Coursework

NIH AIM-AHEAD

- Introduction to Infrastructure, Introduction to Clinical Data Management, Introduction to Good Algorithmic Practices, Introduction to Transnational Team Projects,

The University of the North Carolina at Greensboro

- Mathematical Analysis I and II, Advanced Abstract Algebra, Linear Algebra and Matrix Analysis I and II, Math Software, Ordinary Differential Equations, Partial Differential Equations, Computational Algebra, Computational Number Theory, Numerical Methods, Numerical Analysis, Finite Element Method, Functional Analysis, Advanced Statistics in the Behavioral and Biological Sciences I and II, Doctoral Research Independent Study, Dissertation.

Western Carolina University

- Real Analysis, Design of Experiments, Regression Analysis, Mathematical Modeling, Methods of Applied Mathematics, Introduction to Numerical Analysis, Numerical Analysis.

Mars Hill University

- Applied Linear Algebra, Elementary Probability and Statistics, Probability Theory, Mathematical Statistics, Foundations of Geometry, Differential Equations, Introduction to Computer Science Concepts, Introduction to Computer Science I, Advanced Programming I & II.

References

Dr. Tom Lewis

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Dr. Robert Stolz

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Dr. Diana Bowen (Teaching Reference)

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